DLSL · C2B INTRODUCTION TO DATA SCIENCE · COURSE 42838

Modules 2–3 Study Reviewer

A complete review of the data science & data ethics concepts and the Python programming fundamentals. Everything the module pages cover, reorganised for revision — definitions, facts, syntax, and the points most likely to trip you up.

Module 2 — Data Science | Module 3 — Python | Print-friendly

M2 Module 2 — Introduction to Data Science

10 HOURS · LABELLED "MODULE 1" INSIDE THE COURSE PAGES

LEARNING OUTCOMES

- Define data science and its importance `[CL01]`
- Identify the data science application domains `[CL02]`
- Explain the data science life cycle `[CL03]`

TOPIC OUTLINE

Data Science Defined · Importance of Data Science · Application Domains · Data Science vs. Data Analytics · The Data Science Life Cycle

What is data science?

> **Data science** is an **interdisciplinary field** that focuses on **extracting meaningful insights and knowledge from structured and unstructured data**. It combines techniques from **mathematics, statistics, computer science, and domain expertise** to analyse, interpret, and predict trends or patterns, and is used to solve real-world problems, drive decision-making, and innovate across industries.

- **Structured data** — organised, table-like (rows and columns).

- **Unstructured data** — text, images, audio, video, sensor streams; no fixed schema.
- Closely tied to **Artificial Intelligence (AI)**, which is now used across every field.

> **Trap:** data science is *not* just statistics, not database administration, and not backward-looking reporting/dashboards. The defining phrase is "extracts *insights and knowledge*" from *both* structured *and* unstructured data, and the fourth discipline in the list is **domain expertise** (not software engineering or machine learning).

M2 The data science process

THE CORE STEPS, IN ORDER

1. **Collect** raw data.
2. **Clean and prepare** it for analysis.
3. **Apply statistical and computational methods** to uncover patterns.
4. **Visualise** the results.
5. **Use the insights to inform decisions.**

> **Order matters:** cleaning comes *before* analysis; visualisation comes *after* analysis. Any sequence that visualises first, or analyses before cleaning, is wrong.

WHERE THE DATA COMES FROM (APPLICATION CONTEXT)

Organisations generate huge data volumes every second through **websites, mobile applications, sensors, social media, business transactions, healthcare systems, and Internet of Things (IoT) devices**. Turning that data into knowledge is a competitive advantage.

M2 Why data science matters

THE SEVEN BENEFITS LISTED IN THE MODULE

1	2	3	4
Better decision-making	**Predictive analysis**	**Process optimization**	

Personalized customer experience

5

AI development

6

Scientific research

7

Public policy & governance

Trap: memorise these seven *as worded*. Distractors are usually near-misses (“process *automation*” for process *optimization*, “personalized *marketing*” for personalized *customer experience*) or overreaches the module never claims (“eliminates all bias”, “removes the need for domain experts”, “lowers storage costs”).

DATA SCIENCE VS. DATA ANALYTICS

The module lists this as a topic to distinguish. In general use: **data analytics** examines existing data to explain *what happened and why* (mostly descriptive, on structured data); **data science** is broader — it also builds predictive and prescriptive models, works with unstructured data, and creates new data products and algorithms. *(Check your Lec01 slides for the exact contrast your instructor uses.)*

M2 Data ethics

FROM THE “DATA ETHICS CONCEPTS” PAGE · LEC02

Data ethics is the **system of moral principles that guide the use and management of data**. It covers the ethical issues arising from the **collection, analysis, and dissemination** of data — especially **privacy, security, and the impact on individuals and society**. It concerns the moral obligations governing how data is used, collected, and analysed.

WHY IT MATTERS

- Data is increasingly used in decision-making.
- Unethical handling causes real harm: **privacy infringements, discrimination, and other societal harms**.
- Ethical practice **builds trust** and **protects fundamental human rights**.

> **Trap:** data ethics is a set of *moral principles* — not a law/regulation, not a technical procedure like encryption, and not a certification. It spans the whole data life cycle, not just storage.

GRADED ACTIVITY

GT#1 — Group Activity 1: Case Study on Data Ethics (30 pts). Download the task file, analyse the case, and build a presentation named `gt1-<section>-<groupNo>` (e.g. `gt1-c2b-GRP1.pptx`). One member submits.

M3 Module 3 — Introduction to Python Programming

15 HOURS · LABELLED "MODULE 2" INSIDE THE COURSE PAGES

LEARNING OUTCOMES

- Learn the basics of Python programming.
- Solve problems using Python.

TOPIC OUTLINE

Variables · Constants · Input and Output Operations · Basic Operators in Python · Control Structures

M3 The Python language & its history

FROM "THE PYTHON PROGRAMMING LANGUAGE"

> **Python** is a popular, **high-level, general-purpose** programming language, created with a focus on **code readability**; it lets developers express ideas in fewer lines than C++ or Java.

BRIEF HISTORY

Late 1980s	Dec 1989	Feb 1991	2000	2008
Conceived by **Guido van**	Development begins.	First official release, **Python**	**Python 2** — list comprehensions,	**Python 3** — improved

Rossum at **CWI** (Centrum Wiskunde & Informatica), Netherlands.		**0.9.0**.	garbage collection.	consistency, removed legacy issues.

Designed as a **successor to the ABC language**, focusing on simplicity, readability, and extensibility.

Trap: *development began* Dec 1989; the *first release* (0.9.0) was Feb 1991 — don't swap them. Successor to **ABC** (not Perl, not Modula-3), and a *successor*, not an extension.

STRENGTHS OF PYTHON	KEY FEATURES OF PYTHON
Readable & simple syntax	Interpreted language — no compilation; runs directly from source
Cross-platform (Windows, macOS, Linux, ...)	Object-oriented & procedural support
Extensive libraries (rich standard library + third-party ecosystem)	Dynamic typing — no explicit type declarations
Versatile (web, data science, AI, automation, ...)	High-level — abstracts complex details
Strong community support	Extensible & embeddable — integrates with C/C++
	Automatic memory management — built-in garbage collection

M3 Variables, input & output

FROM “PYTHON BASICS” AND “VARIABLES, DATA TYPES AND OPERATORS”

STRUCTURE OF A PYTHON PROGRAM

- **Import statements** — include external Python modules.
- **Functions and classes** — the building blocks of code.
- **Statements and expressions** — executable instructions.
- **Comments** — descriptions and explanations inside the code.

OUTPUT — PRINT()

Used for output in various formats. Prints single or multiple values; separate multiple values with commas.

```python
print("Hello, World!")

s = "Brad"
print(s)

s = "Anjelina"
age = 25
city = "New York"
print(s, age, city)    # Anjelina 25 New York
```

INPUT — INPUT()

Takes user input. **By default it returns the input as a string** — convert it (e.g. `int(...)`, `float(...)`) if you need a number.

```python
name = input("Enter your name: ")
print("Hello,", name, "! Welcome!")
```

VARIABLES

> A **variable** is a **name assigned to a value**, used to store data that can be referenced and manipulated while the program runs. Unlike Java, Python needs **no explicit type declaration** — the type is **inferred from the value assigned** (dynamic typing).

```python
x = 5
name = "Alex"
print(x)
print(name)
```

RULES FOR NAMING VARIABLES

1. Names can contain **letters, digits, and underscores** (`_`).
2. The **first character cannot be a digit**.
3. Names are **case-sensitive** — `myVar` and `myvar` are different.
4. **Keywords** (`if`, `else`, `for`, ...) cannot be used as names.

> **Trap:** each of these four is a favourite true/false and MCQ target. Underscores *are* allowed; the first character *cannot* be a digit; names *are* case-sensitive; keywords are *not* allowed.

M3 Data types & operators

PAGE HEADINGS ONLY — THIS IS THE STANDARD CONTENT FOR THAT TOPIC (CONFIRM AGAINST LEC03)

COMMON BUILT-IN DATA TYPES

CATEGORY	TYPES	EXAMPLE
Numeric	`int`, `float`, `complex`	`5`, `3.14`, `2j`
Text	`str`	`"Alex"`
Boolean	`bool`	`True`, `False`
Sequence	`list`, `tuple`, `range`	`[1, 2, 3]`, `(1, 2)`
Mapping	`dict`	`{"age": 25}`
Set	`set`, `frozenset`	`{1, 2, 3}`
None	`NoneType`	`None`

OPERATOR GROUPS

GROUP	OPERATORS
Arithmetic	`+` `-` `*` `/` `%` `//` `**`
Comparison	`==` `!=` `>` `<` `>=` `<=`
Assignment	`=` `+=` `-=` `*=` `/=` `//=` `**=` `%=`
Logical	`and` `or` `not`
Identity	`is` `is not`
Membership	`in` `not in`
Bitwise	`&` `|` `^` `~` `<<` `>>`

> **Note:** the module page lists “Data Types in Python” and “Operators in Python” as headings with the detail in the Lec03 slides / linked GeeksforGeeks & Cisco *Python Essentials 1* resources. The tables above are the standard set — verify the exact list your instructor tests.

M3 Control structures

FROM “CONTROL STRUCTURES IN PYTHON”

> **Control structures** govern the **flow of execution**. Conditional statements run different blocks depending on whether a condition is `True` or `False`; loops repeat a block until a condition is met or a sequence is exhausted.

CONDITIONAL (IF) STRUCTURES — SEVEN FORMS

1. **If statement**
2. **Short-hand If**
3. **If-else statement**
4. **If-elif-else statement**
5. **Nested if-else**
6. **Conditional expression** (ternary operator)
7. **Match-case statement**

LOOP STATEMENTS

- **For loop** — iterates over sequences.
- **While loop** — runs while a condition holds.
- **Nested loops** — a loop inside another loop.

```python
# if-elif-else
if score >= 90:
    grade = "A"
elif score >= 75:
    grade = "B"
else:
    grade = "C"

# ternary / conditional expression
status = "pass" if score >= 75 else "fail"

# for loop over a sequence
for item in ["a", "b", "c"]:
    print(item)

# while loop
n = 3
while n > 0:
    print(n)
    n -= 1
```

GRADED ACTIVITY

GT#2 (Q1) — Python Basics in Codechum (30 pts). Completed on the Codechum platform.

⋆ Quick-recall cheat sheet

DS DEFINITION	DS DISCIPLINES	DS PROCESS	DS BENEFITS
Interdisciplinary field insights & knowledge from structured + unstructured data	**Math · Statistics · CS · Domain expertise**	**Collect → Clean → Analyse → Visualise → Decide**	**7** decisions, prediction, optimization, personalization, AI, research, policy

DATA ETHICS	PYTHON IS	CREATOR	DEV BEGAN
	High-level, general-purpose	**Guido van Rossum** at CWI, Netherlands	**December 1989**

Moral principles
use & management of data; privacy, security, society

readability-focused, interpreted, dynamically typed

FIRST RELEASE
Feb 1991
Python 0.9.0

PY 2 / PY 3
2000 / 2008

SUCCESSOR TO
ABC language

INPUT() RETURNS
a string (by default)

PRINT() MULTIPLE
separate with commas

TYPING
Dynamic
type inferred from the value; no declaration

NAMING RULES
letters/digits/_ · no leading digit · case-sensitive · no keywords

IF FORMS
7
if, short-hand, if-else, if-elif-else, nested, ternary, match-case

LOOPS
For · While · Nested

MEMORY
Automatic
built-in garbage collection

⋆ Key terms glossary

TERM	MEANING
Data science	Interdisciplinary field extracting insights & knowledge from structured and unstructured data.
Structured data	Data organised in a fixed schema (rows/columns).
Unstructured data	Data with no fixed schema — text, images, audio, video, sensor data.
Domain expertise	Subject-area knowledge; the 4th ingredient of data science.
Data science life cycle	Collect → clean/prepare → apply methods → visualise → inform decisions.
Predictive analysis	Using data/models to forecast future trends or outcomes.
IoT	Internet of Things — networked devices/sensors that generate data.
Data ethics	System of moral principles guiding the use and management of data.
High-level language	Abstracts hardware detail; closer to human language.
Interpreted language	Runs directly from source code, no separate compile step.
Dynamic typing	Variable type is inferred from the assigned value at run time.
Garbage collection	Automatic reclaiming of unused memory.
Variable	A name bound to a value.
Keyword	Reserved word (`if`, `for`, …); cannot be a variable name.
Statement / expression	Executable instruction / a fragment that evaluates to a value.
Ternary / conditional expression	`A if cond else B` — one-line conditional.
ABC	The language Python was designed to succeed.
CWI	Centrum Wiskunde & Informatica (Netherlands) — where Python began.

★ Common exam traps

1. **DS ≠ statistics / databases / reporting.** It's interdisciplinary, works with structured *and* unstructured data, and produces insight, not just reports.
2. **The 4th discipline is "domain expertise."** Not software engineering, not machine learning, not project management.
3. **Process order:** clean *before* analyse; visualise *after* analyse.
4. **The 7 benefits** must be recalled by exact wording — beware near-miss distractors.
5. **Data ethics = moral principles**, not a law, encryption process, or certificate; spans collection, analysis, *and* dissemination.
6. **Python timeline:** conceived late 1980s · dev began **Dec 1989** · first release **Feb 1991 (0.9.0)** · Py2 **2000** · Py3 **2008**.
7. **Successor to ABC** — a *successor*, not an extension; not Perl or Modula-3.
8. `input()` **returns a string** by default — no automatic number conversion.
9. `print()` can take multiple values separated by commas.
10. **Dynamic typing:** no declaration; type inferred from the value. Values are *not* all stored as strings; no type hint is required.
11. **Naming rules:** underscores allowed · no leading digit · case-sensitive · no keywords. Each clause is a separate trap.
12. **Seven IF forms** including the ternary ("conditional expression") and **match-case**.

Compiled from the Canvas module pages of *C2B Introduction to Data Science* (course 42838), Modules 2–3 (Data Science and Python), for personal revision. Items marked "confirm against Lec03" are standard supplementary content where the module page gave only a heading. Pair this with the companion **Data Science Review Quiz** to self-test.